

Anoop Gupta

AI/ML Engineer & Full-Stack Developer

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<https://github.com/AnoopG7> | <https://anoopg-portfolio.vercel.app/>

PROFESSIONAL SUMMARY

AI/ML Engineer and Full-Stack Developer with hands-on experience building production-grade AI systems, RAG architectures, multi-agent workflows, and scalable SaaS platforms using FastAPI, React, TypeScript, PostgreSQL, and modern LLM tooling. Skilled in LLM applications, backend architecture, cloud deployment, and production reliability. Passionate about building intelligent products with strong backend architecture, performance optimization, and production reliability.

Skills

AI / Machine Learning - Python, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Hugging Face, OpenAI API, LangChain, RAG Systems, Multi-Agent Workflows, Prompt Engineering, Model Evaluation, Semantic Search, Vector Embeddings, pgvector

Backend Development - FastAPI, Node.js, Express.js, REST APIs, WebSockets, Socket.io, JWT Authentication, SQLAlchemy Async ORM, Alembic, Async Processing, API Integration

Frontend Development - React, Next.js, TypeScript, JavaScript, Tailwind CSS, Zustand, React Query, Shadcn UI, MUI, React Hook Form

Databases - PostgreSQL 16, MongoDB, MySQL, Supabase, Firebase, Database Schema Design

DevOps - Docker, Docker Compose, Kubernetes, Git, GitHub, GitHub Actions, CI/CD, Linux, Jenkins, Nginx, Swagger/OpenAPI, Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana)

System Design - Distributed Systems, Scalable Backend Architecture, Microservices, Event-Driven Architecture, Message Queues, Caching, Load Balancing, API Design

Cloud – Terraform, AWS (EC2, S3, IAM, VPC, Auto Scaling, Elastic Load Balancer, CloudWatch)

Experience

AI Engineer Intern | Shovihini Tech LLP

- Built AI-powered applications using LLMs, RAG pipelines, vector embeddings, and NLP workflows
- Developed scalable backend services for AI inference and data processing using Python and FastAPI
- Improved retrieval accuracy, latency, and response quality through prompt and pipeline optimization

Web Development Intern | CodeDev Verse

- Developed responsive full-stack web applications using modern frontend and backend technologies
- Built APIs, integrated databases, and implemented authentication workflows
- Contributed to debugging, testing, deployment, and performance optimization

FEATURED PROJECTS

AI Lead Gen Platform | AI-Powered Lead Generation SaaS

Tech Stack: FastAPI, PostgreSQL, React, Docker, OpenAI, Meta Graph API

- Built a deterministic AI agent using raw OpenAI function calling with 8 tools and a 10-turn autonomous execution loop
- Developed async FastAPI backend with custom JWT auth, CRM dashboard, and 16-table PostgreSQL schema
- Reduced inference costs by ~60% using Groq Llama-3 with 429 retry backoff, and JSON recovery

TutorX | AI-Powered RAG Tutoring System

Tech Stack: FastAPI, React, Supabase, pgvector, Groq

- Built a RAG tutor using NCERT textbooks for curriculum-aligned, context-aware explanations
- Designed 3-stage retrieval (pgvector search, query reformulation, chunk enrichment), reducing API calls from 14+ to 1
- Implemented PDF ingestion, embeddings generation, authentication, and Telegram bot integration

TaskFlow | AI-Powered Proactive Task Management

Tech Stack: FastAPI, React, Supabase, Telegram Bot

- Built a multi-action AI agent capable of executing 10 task operations atomically in a single LLM call
- Developed self-improving time estimation with bias tracking across the last 30 completed tasks
- Built APScheduler workflows for daily briefings, hourly alerts, and multi-channel task management

AI-Powered LMS | MERN + AI Integration

Tech Stack: Node.js, Express, MongoDB, React, Socket.io

- Built AI-powered flashcard and quiz generation from uploaded study materials using LLM workflows
- Developed real-time collaborative chat with Socket.io and adaptive quiz analytics dashboards
- Designed modular backend architecture with secure file-processing pipelines and authentication

ADDITIONAL PROJECTS

Building Management System, WhatsApp AI Assistant, Telegram AI Bots, Real-Time Chat Applications, Full-Stack SaaS Tools

More projects: Portfolio → anoopg-portfolio.vercel.app

EDUCATION

Pursuing Bachelor of Technology (B.Tech) in Computer Science Engineering - AI/ML Specialization

ITM Skills University, Navi Mumbai, India | Graduation: 2028 | CGPA: 9.5/10
